Regular-cone Jacobi identities and controlled scalar boundary response

Ricardo Maldonado
Independent Researcher
ORCID: 0009-0009-3937-6527

11 October 2026

Abstract

We study a static Einstein–scalar boundary problem for two identical regular interiors joined at a de Sitter brane. A regular-cone Jacobi identity gives an exact curvature factor in the derivative of the geometric junction residual. The existing weighted contraction conditions imply strict bulk convexity; this in turn proves strict residual monotonicity and removes a previously restrictive sufficient inequality for local branch uniqueness. A Riccati comparison sharpens the endpoint estimate. For the specified exact-decimal polynomial model, exact arithmetic verifies all remaining sufficient conditions for every $0 < \delta \leq 0.002$, together with explicit scalar and curvature remainders and lower curvature relative to a separate model with scalar-independent detuning. Uniqueness is restricted to the stated tube and $T \geq 1/k$. The note includes all conditions and the mathematical proof.

Research and review status. This is an AI-assisted mathematical research note with internal mathematical and computational review; it has not received external peer review. OpenAI Codex assistance was used for derivation, checking, programming, and drafting. The author attribution does not by itself assert a personal rerun of every computation. The results are conditional static mathematical theorems under the explicit hypotheses below. They do not establish dynamical stability, a cosmological origin mechanism, or evidence for a higher-dimensional origin of the Big Bang. No global priority or novelty claim is made.

Source preservation. The accompanying frozen proof source in the capsule's `theory` directory is the exact frozen proof source. This typeset note preserves its complete mathematical body, including all eleven sufficient inequalities and the registered-model corollary. The source hash and reproducible compilation receipt accompany the TeX and PDF.

11 October 2026. Local mathematical continuation for review. This file does not modify or replace the accepted project files. Its conclusions concern the static Einstein–scalar boundary problem and its specified small Dirichlet tube. No global stability, dynamical, cosmological, or external novelty claim is made.

1. Conclusions

The sufficient "strict downward shell crossing'' inequality in Section 6 of the 8 October uniform branch proof can be deleted. Under its other construction and scalar-junction

hypotheses, the metric residual along the scalar-junction locus satisfies the exact sign estimate

$$\frac{d}{dT}\left[a(T;\beta(T)) - \frac{\sigma(\beta(T))}{6}\right] \le -\rho(T;\beta(T))^{-2} < 0. \tag{1}$$

The first inequality is strict for every nonconstant scalar interior. The argument also applies to the derivative norm with an arbitrary positive scale s, with its correctly propagated constants given below. In particular, the conservative former slope constant is unnecessary; it was an artifact of estimating two terms separately that cancel at leading order.

A separate comparison estimate improves the linear Dirichlet-to-Neumann remainder to

$$0 \le \lambda - \frac{F_\lambda'(T)}{F_\lambda(T)} \le \frac{\lambda}{k(w-3k)} H_0(T)^2. \tag{2}$$

Consequently the constant called C_H in the baseline proof may be replaced by

$$C_H^{\text{new}} = \frac{\lambda}{k(w-3k)}. \tag{3}$$

Neither improvement changes the boundary expansion coefficients. Their utility is a larger interval and smaller *proved upper bounds* on the remainders, when the remaining sufficient inequalities are evaluated.

2. Equations, variables, and normalization

Let $W \in C^4([-r,r])$, $f \in C^3([-r,r])$,

$$W(0) = 3k, \quad W'(0) = 0, \quad W''(0) = w > 4k > 0, \quad f(0) = f_0 > 0,$$

and set

$$U = \frac{1}{2}(W')^2 - \frac{2}{3}W^2, \quad m^2 = U''(0) = w(w-4k), \quad \lambda = w - 4k, \quad \sigma = 2W + \delta f.$$

Primes on U, W, f, σ mean scalar derivatives; primes on radial quantities mean derivatives in y. Put

$$p = \eta', \qquad a = \rho'/\rho, \qquad \mathcal{H} = \rho^{-2}.$$

For $y > 0$, the regular interior obeys

$$p' = U'(\eta) - 4ap, \qquad a' = -\mathcal{H} - \frac{p^2}{3}, \qquad a^2 = \mathcal{H} + \frac{p^2}{12} - \frac{U(\eta)}{6}. \tag{4}$$

Equivalently,

$$a' = -a^2 - \frac{p^2}{4} - \frac{U(\eta)}{6}. \tag{5}$$

The cone data are $\rho(0) = 0, \rho'(0) = 1, p(0) = 0$. Its sole scalar datum is $c = \eta(0)$. Radius normalization is fixed: it is not an extra Jacobi parameter. At the endpoint, $\beta = \eta(T)$ and $H^2 = \mathcal{H}(T)$.

The weighted construction supplies a C^1 family of regular interiors at each finite $T > 0$, with $\eta(T; T, \beta) = \beta$. All partial derivatives in β below hold T fixed. The scalar junction is

$$g(T, \beta) = p(T; T, \beta) + \frac{1}{2}\sigma'(\beta) = 0, \quad g_\beta = p_\beta + \frac{1}{2}\sigma'' > 0. \tag{6}$$

In the baseline sufficient conditions, $g_\beta \geq w/2$.

3. Cone Jacobi fields and their endpoint identities

Differentiate the cone solution with respect to c, at fixed y:

$$h = \eta_c, \quad q = p_c = h', \quad v = (\log \rho)_c, \quad b = a_c = v'.$$

For finite regular segments $\rho > 0$, cone parameter dependence and ordinary ODE parameter dependence justify these derivatives. Their equations are

$$q' = U''h - 4aq - 4pb, \qquad b' = 2\mathcal{H}v - \frac{2}{3}pq, \qquad v' = b. \tag{7}$$

The differentiated constraint is

$$2ab = -2\mathcal{H}v + \frac{pq - U'h}{6}. \tag{8}$$

Define the two Jacobi combinations

$$j = b + \frac{ph}{3}, \qquad \mathcal{D} = hp' - pq. \tag{9}$$

Direct substitution of (4), (7), and (8) gives

$$j' + 4aj = -2\mathcal{H}v, \tag{10}$$

$$\mathcal{D}' + 4a\mathcal{D} = 4p\big(\mathcal{H}h + pj\big). \tag{11}$$

For clarity, (8) implies $pq - p'h = 12aj + 12\mathcal{H}v$. Substituting this into $j' = 2\mathcal{H}v + (p'h - pq)/3$ proves (10). For (11), differentiate $hp' - pq$, cancel $qp' - p'q$, and use $p'' = U''p - 4a'p - 4ap'$. The result is $-4a\mathcal{D} - 4a'ph + 4p^2b$, which is (11).

The regular cone expansions, including their parameter derivatives, are

$$\begin{aligned} &\rho = y - U(c)y^3/36 + O(y^5), && p = U'(c)y/5 + O(y^3), \\ &h = 1 + U''(c)y^2/10 + O(y^4), && q = U''(c)y/5 + O(y^3), \\ &v = -U'(c)y^2/36 + O(y^4), && b = -U'(c)y/18 + O(y^3). \end{aligned} \tag{12}$$

Only the displayed leading terms and smaller remainders are needed. They follow from the regular cone integral equations and their differentiated equations. Thus

$$j = U'(c)y/90 + O(y^3), \qquad \mathcal{D} = U'(c)/5 + O(y^2). \tag{13}$$

In particular, *both* $\rho^4 j$ and $\rho^4\mathcal{D}$ tend to zero at the cone; $\mathcal{D}$ itself need not tend to zero. Integrating (10)–(11) with their proper cone boundary terms yields

$$j(y) = -2\rho(y)^{-4} \int_0^y \rho(t)^2 v(t)\, dt, \tag{14}$$

$$\mathcal{D}(y) = 4\rho(y)^{-4} \int_0^y \rho(t)^4 p(t)[\mathcal{H}(t)h(t) + p(t)j(t)]\,dt. \tag{15}$$

The second integrand is $O(t^3)$ at zero, so no singular boundary term is hidden in this formula.

Whenever $h(T) \neq 0$, the cone-to-endpoint map is locally invertible and

$$p_\beta(T) = q(T)/h(T), \quad a_\beta(T) = b(T)/h(T), \quad Q(T) := a_\beta(T) + p(T)/3 = j(T)/h(T). \tag{16}$$

The regular Dirichlet family already ensures this nondegeneracy: writing $c = c(T, \beta)$, differentiation of its endpoint condition gives $h(T)c_\beta = 1$. The positivity proof below additionally gives $h(T) \geq 1$.

Changing T at fixed β gives $c_T = -p/h(T)$, and therefore

$$\partial_T p|_\beta = p' - pp_\beta = \mathcal{D}/h(T), \quad \partial_T a|_\beta = -H^2 - p^2/3 - pa_\beta. \tag{17}$$

All statements here concern finite T; no implicit-function theorem at $T = \infty$ is used.

Along (6), $\beta_T = -(\mathcal{D}/h(T))/g_\beta$. For $\mathscr{R}(T) = a(T; T, \beta(T)) - \sigma(\beta(T))/6$, use $\sigma' = -2p$ to obtain the exact cancellation

$$\mathscr{R}_T = -H^2 + Q(T)(\beta_T - p). \tag{18}$$

This identity holds everywhere on the scalar-junction locus, not merely at a metric-junction root.

4. Convexity gives the signs needed in (18)

Assume the scalar stays in an interval containing zero on which $U'' > 0$, and assume $\rho > 0$ away from the cone. If $c \neq 0$, let $\epsilon = \operatorname{sign} c$. The scalar divergence equation

$$(\rho^4 p)' = \rho^4 U'(\eta)$$

and $U'(0) = 0$ show that $\epsilon p > 0$, $\epsilon U'(\eta) > 0$, and $\epsilon\eta > 0$ for $y > 0$ on the segment: η starts with sign c, moves away from zero, and a first failure of that assertion contradicts the integral formula for p.

Put $z = -\epsilon b$. Differentiation of (5), rather than of the constraint form of a', gives the particularly simple positive system

$$(\rho^4 q)' = \rho^4(U''h + 4|p|z), \qquad (\rho^2 z)' = \rho^2(|p|q/2 + |U'|h/6), \qquad h' = q. \tag{19}$$

The cone terms in the first two integrated equations vanish. The coefficients coupling different unknowns are nonnegative. By (12), for sufficiently small positive y all of h, q, z are positive. Start at any such point; the ordinary cooperative linear system preserves nonnegativity, as can also be seen by a first-zero argument after applying the displayed integrating factors. Thus on the full segment

$$h \geq 1, \qquad q > 0, \qquad z > 0.$$

This argument avoids assigning arbitrary zero data to a singular Jacobi equation. The strict center coefficients fix the regular solution. Since $v(0) = 0$, $v' = b = -\epsilon z$ implies $\epsilon v < 0$. Equation (14) now gives $\epsilon j > 0$, hence $pj > 0$. Equation (15) then gives $\epsilon\mathcal{D} > 0$.

It follows from (16)–(17) that Q has the sign of p and $\partial_T p|_\beta$ also has the sign of p. With $g_\beta > 0$, β_T has the opposite sign. In particular

$$Q(\beta_T - p) < 0 \quad (c \neq 0),$$

which proves (1), with strict first inequality. This proof requires no sign assumption on σ'' beyond the actual positive scalar-junction derivative g_β. If $c = 0$, the interior is the exact scalar-free cone, $p = b = v = j = 0$, and (18) gives equality $\mathscr{R}_T = -H^2$.

5. Strict convexity is already implied by the construction

This step is essential: Section 4 adds no convexity hypothesis to the uniform theorem's surviving sufficient conditions. Let

$$M_2 = \sup_{[-r,r]} |U''|, \quad M_3 = \sup_{[-r,r]} |U'''|, \quad L_1 = M_3/2,$$

and choose $\nu = \lambda/2$, $\mu_\nu = \nu(\nu + 4k)$, $D = m^2 - \mu_\nu > 0$. With arbitrary derivative scale $s > 0$, the relevant weighted norm and constants are

$$\begin{aligned} \|\eta\|_{T,s} &= \max\{\sup|\eta|/\omega_T, \sup|\eta'|/(s\omega_T)\}, \quad \omega_T = F_\nu/F_\nu(T), \\ M &= \max(1, \lambda/s), & B &= 2M, \\ C_G &= \max\{D^{-1}, (1 + m^2/D)/(4ks)\}, & C_F &= s^2/4 + M_2/12, \\ C_N &= L_1 B^2 + 4C_F(s/k)B^3 b, & L_N &= 2L_1 Bb + 20C_F(s/k)B^2 b^2. \end{aligned} \tag{20}$$

The tube conditions include $Bb \le r$, $\|\eta\|_{T,s} \le B|\beta| \le Bb$, and $q_{\text{contr}} := C_G L_N \le 1/2$. Since $C_G \ge D^{-1}$ and $L_N \ge M_3 Bb$,

$$M_3 Bb \le D/2, \quad U''(x) \ge m^2 - M_3|x| \ge m^2 - D/2 = (m^2 + \mu_\nu)/2 > 0 \quad (|x| \le Bb). \tag{21}$$

Thus every interior in the tube satisfies the strict convexity used above.

There is also a useful metric consequence. Since $U'(0) = 0$, strict convexity on this interval implies $U(\eta) \ge U(0) = -6k^2$. Thus the Volterra source $\mathcal{F} = p^2/4 + [U(\eta) + 6k^2]/6$ is nonnegative along the actual solution. The already established positivity of $R = \rho/r_0$, together with $R' = -\int_0^y \partial_y K(y,t)\mathcal{F}(t)R(t)\,dt$ and $\partial_y K \ge 0$, implies $R' \le 0$. Since $R(0) = 1$,

$$0 < R(y) \le 1, \qquad H_0(T)^2 = R(T)^2 H^2 \le H^2. \tag{21a}$$

This sign improvement is only asserted on fixed-point solutions; the contraction estimates on arbitrary candidate functions retain their original absolute-value bounds.

The norm propagation in (20) follows by replacing $|p| \le kB|\beta|\omega_T$ by $|p| \le sB|\beta|\omega_T$. The metric source has bound $C_F B^2 \beta^2 \omega_T^2$, its logarithmic derivative still has geometric denominator k, and the product $(a - a_0)p$ therefore acquires s/k. The physical derivative Green bound is unchanged, and division by s gives exactly the displayed C_G. In particular setting $s = k$ recovers the original proof.

Keep every sufficient condition used for the positive metric, Dirichlet contraction, scalar-junction monotonicity, and the opposite signs of the metric residual at $T = 1/k$ and $T \to \infty$. Delete only the former condition $C_{\text{slope}}\bar{\delta} \le a_1/4$. The remaining construction proves a scalar-junction root $\beta(T, \delta)$ for all $T \ge 1/k$, and Section 4 proves that its metric residual is strictly decreasing. Its opposite endpoint signs therefore give exactly one metric root in the same tube. The strictly negative derivative supplies the ordinary C^1 implicit-function theorem at every positive δ, just as in the baseline proof.

6. A quantitative alternative that does not use sign arguments

This independent estimate isolates the curvature factor even without convexity. It can be useful for other settings. Let $R = \rho/r_0$, $r_0 = \sinh(ky)/k$, and let J be the uniform bound $\int_0^T \omega_T^2 \le J$ used in the construction. Suppose its Volterra smallness bound is $\varepsilon_V = C_F B^2 b^2 J/(2k) \le 1/4$. Then $2/3 \le R \le 4/3$, and differentiation of $R = 1 - \mathcal{V}(\mathcal{F}R)$, where $\mathcal{F} = p^2/4 + (U + 6k^2)/6$, yields

$$\|R_\beta\|_\infty \le \frac{16}{9}\frac{C_F B^2 J}{k}|\beta|.$$

Indeed $|\mathcal{F}_\beta| \le 2C_F B^2|\beta|\omega_T^2$, the inverse Volterra norm is at most $4/3$, the metric factor at most $4/3$, and the kernel at most $1/(2k)$. Hence, for the endpoint-normalized variation $v_\beta = \partial_\beta \log \rho$,

$$\|v_\beta\|_\infty \le C_v|\beta|, \qquad C_v = \frac{8C_F B^2 J}{3k}. \tag{22}$$

If $a \ge a_{\min} > 0$, then $\int_0^T \rho^2 \le \rho(T)^2/(2a_{\min})$. Dividing (14) by $h(T)$ therefore proves the explicit curvature-factor bound

$$|a_\beta + p/3| \le \frac{C_v}{a_{\min}} H^2|\beta|. \tag{23}$$

The existing metric estimate permits

$$a_{\min} = k - C_F B^2 b^2/k \ge k/2.$$

With the baseline constants

$$\begin{aligned} C_P &= sC_G B[2L_1 B + 20C_F(s/k)B^2 b], \\ \bar{a} &= k\coth(1) + C_F B^2 b^2/k, \\ C_{Tp} &= M_2 + 4\bar{a}sB + sB(\lambda + C_P b), \\ C_{T\beta} &= 2C_{Tp}/w, \end{aligned}$$

one has $|\beta_T - p| \le (C_{T\beta} + sB)|\beta|$. For $|\beta| \le L\delta$, (18) gives

$$\mathscr{R}_T \le -H^2(1-\theta), \qquad \theta = \frac{C_v}{a_{\min}}(C_{T\beta} + sB)L^2\bar{\delta}^2. \tag{24}$$

Thus $\theta < 1$ is a sufficient explicit substitute for the former slope condition, independently of Sections 4–5. For rational computation, $\coth(1) < 4/3$ gives a rational upper replacement for $\bar{a}$.

7. Improved linear endpoint comparison

Let F_λ be the positive regular solution of

$$F_\lambda'' + 4a_0 F_\lambda' - m^2 F_\lambda = 0, \quad F_\lambda(0) = 1,\ F_\lambda'(0) = 0, \quad a_0 = k\coth(ky).$$

Its logarithmic derivative $z = F_\lambda'/F_\lambda$ obeys $z' = m^2 - 4a_0 z - z^2$ and $0 \le z \le \lambda$, by the positive divergence equation and the first-crossing test at $z = \lambda$. Put

$$d = a_0 - k > 0, \quad e = \lambda - z \ge 0, \quad E = e/d.$$

The exact equations are

$$d' = -d(d+2k), \quad e' = -(w+z)e + 4dz, \quad E' = 4\lambda - 2(w-3k)E + dE(E-3). \tag{25}$$

At the regular cone $d \sim 1/y$, $z = O(y)$, so $E \to 0$. Let $C = 2\lambda/(w-3k)$. Since $w > 4k$, one has $0 < C < 2$. At the barrier $E = C$, the first two terms of (25) cancel while $dC(C-3) < 0$. Starting at a sufficiently small positive radius, the usual first-crossing argument therefore proves $0 \le E < C$ for every finite positive radius. Since $H_0^2 = a_0^2 - k^2 = d(a_0+k)$ and $a_0 + k \ge 2k$, this yields (2).

At a nonlinear endpoint, (21a) gives $H_0^2 = R(T)^2 H^2 \le H^2$. The unchanged nonlinear endpoint bound

$$|p(T) - z(T)\beta| \le C_D \beta^2, \qquad C_D = sC_G C_N,$$

therefore gives

$$|p(T) - \lambda\beta| \le C_H^{\text{new}} H^2 |\beta| + C_D \beta^2, \tag{26}$$

with (3). If the sign information (21a) is unavailable in a different problem, the absolute-value metric bound $R \le 4/3$ still gives the weaker constant $16\lambda/[9k(w-3k)]$. Unlike the former exponential estimate, (26) holds at every $T > 0$; the branch construction may still use $T \ge 1/k$ for its other conditions.

8. Remainder constants and scope

Choose any $L \ge |f'(0)|/w$, and use the notation

$$a_1 = kf_0/3, \quad d_0 = 2(w-2k), \quad \alpha = -f'(0)/(2d_0),$$

The choice $L = |f'(0)|/w$ is valid because $g(T,0) = \delta f'(0)/2$ and $g_\beta \ge w/2$ imply $|\beta(T,\delta)| \le |f'(0)|\delta/w$. The baseline extra $1/w$ in L is not needed. If $f'(0) = 0$, then $\beta = 0$ is exactly the unique scalar-junction root, and any independently positive Dirichlet tube radius b may be used; taking $L = 0$ is harmless in the remainder formulas.

Define the endpoint functions

$$A(x) = W(x)f(x)/9 - W'(x)f'(x)/12, \quad B_0(x) = f(x)^2/36 - f'(x)^2/48.$$

With $W_3 = \sup|W'''|$, $F_2 = \sup|f''|$, all on $[-r, r]$, the improved sufficient remainder constants are

$$K_\eta^{\text{new}} = \frac{C_H^{\text{new}}(3a_1/2)L + (C_D + W_3/2)L^2 + F_2 L/2}{d_0},$$

$$K_H^{\text{new}} = |A'(0)|K_\eta^{\text{new}} + \frac{1}{2}\sup|A''|L^2 + \sup|B_0'|L.$$

They give the same uniform estimates

$$|\beta - \alpha\delta| \le K_\eta^{\text{new}} \delta^2,$$

$$\left| H^2 - a_1\delta - \left[\frac{f_0^2}{36} - \frac{kf'(0)^2}{24(w-2k)}\right]\delta^2 \right| \le K_H^{\text{new}} \delta^3.$$

For polynomial endpoint functions $A(x) = \sum_i A_i x^i$ and $B_0(x) = \sum_i B_i x^i$, an additional exact refinement is available. For a fixed interval endpoint $\bar\delta$, put

$$E_* = \min\{L, |\alpha| + K_\eta^{\text{new}}\bar\delta\},$$

$$K_H^{\mathrm{poly}} = |A_1| K_\eta^{\mathrm{new}} + \sum_{i\ge 2} |A_i| E_*^i \bar\delta^{i-2} + \sum_{i\ge 1} |B_i| E_*^i \bar\delta^{i-1}. \tag{27}$$

Indeed $|\beta| \le E_*\delta$, and the exact shell constraint is $H^2 = \delta A(\beta) + \delta^2 B_0(\beta)$. Subtract its constant and linear-in-β terms, use $|\beta - \alpha\delta| \le K_\eta^{\mathrm{new}}\delta^2$, and bound each remaining polynomial term. This proves the same curvature remainder with $K = \min(K_H^{\mathrm{new}}, K_H^{\mathrm{poly}})$. No series truncation is involved.

For the distinct comparator with scalar-independent detuning $f \equiv f_0$, its curvature is exactly $a_1\delta + f_0^2\delta^2/36$. Strict lower curvature therefore follows only when the additionally evaluated margin $kf'(0)^2/[24(w-2k)] - K\bar\delta > 0$ is positive. This final sign claim must not be inferred from existence or monotonicity alone.

9. Complete sufficient-condition checklist and construction summary

Here is a complete way to apply this continuation without having to infer which old inequality survives. Choose $r, b, s, \bar\delta > 0$ and $L \ge |f'(0)|/w$, with the model assumptions of Section 2. Use (20), the definition of C_P in Section 6, and

$$C_0 = 4k\coth(1) + \nu, \quad \zeta = \frac{\mu_\nu[1 - \exp(-C_0/k)]}{C_0}, \quad J = \frac{2}{k} + \frac{1}{2\zeta}.$$

Define derivative bounds on $[-r, r]$:

$$W_2 = \sup|W''|, \quad W_3 = \sup|W'''|, \quad F_2 = \sup|f''|, \quad F_{\mathrm{abs}} = \sup|f|, \quad C_K = L\sup|A'| + \sup|B_0|.$$

The following eleven conditions suffice; the sixth is retained as a strict first-stage feasibility condition, although the scalar-junction derivative bound is what is directly used for each δ:

Condition	Required inequality
Field tube	$Bb \le r$
Metric Volterra inverse	$C_F B^2 b^2 J/(2k) \le 1/4$
Scalar contraction	$C_G L_N \le 1/2$
Scalar ball preservation	$C_G C_N b \le M$
Positive metric derivative	$C_F B^2 b^2/k \le k/2$
Scalar derivative feasibility	$(C_P + W_3)b < w/2$
Scalar root inside tube	$L\bar\delta \le b$
Scalar-junction derivative	$(C_P + W_3)b + \bar\delta F_2/2 \le w/2$
Positive reduced curvature	$C_K\bar\delta \le a_1/2$
Positive residual at $Y = 1/k$	$(3a_1/2)\bar\delta < (9/16)H_0(Y)^2$
Positive shell tension	$W_2 b^2/6 + \bar\delta F_{\mathrm{abs}}/6 \le k/2$

An upper bound for any displayed nonnegative constant is sufficient. In particular, wholly rational replacements are

$$\zeta \ge \frac{(15/16)\mu_\nu}{16k/3 + \nu}, \qquad J \le \frac{2}{k} + \frac{16k/3 + \nu}{(15/8)\mu_\nu}, \qquad H_0(1/k)^2 > k^2/2.$$

They follow from $\coth(1) < 4/3$, $C_0/k > 4$, $\exp(4) > 16$, and $\sinh(1)^2 < 2$. Thus the residual condition can be checked by the stronger rational inequality $(3a_1/2)\bar{\delta} < 9k^2/32$.

For completeness, the construction behind this checklist has the following steps. For $\omega_T = F_\nu/F_\nu(T)$, positivity of $(r_0^4 F_\nu')' = \mu_\nu r_0^4 F_\nu$ gives $0 < \omega_T \le 1$, $0 \le \omega_T' \le \nu\omega_T$, and the large-radius lower bound $F_\nu'/F_\nu \ge \zeta$ for $y \ge 2/k$ gives $\int_0^T \omega_T^2 \le J$. The scalar Green inverse of $\mathcal{L}_m v = v'' + 4a_0 v' - m^2 v$, with regular center data and $v(T) = 0$, has weighted norm at most C_G. Its scalar bound follows by the maximum principle from $\mathcal{L}_m \omega_T = -D\omega_T$; integration of its divergence equation gives the derivative bound in (20).

For any trial scalar in $\|\eta\|_{T,s} \le B|\beta|$, the metric is the unique Volterra solution

$$R(y) = 1 - \int_0^y \frac{\sinh(k(y-t))\sinh(kt)}{k\sinh(ky)} \mathcal{F}(t) R(t)\, dt.$$

The kernel is between zero and $1/(2k)$, and its radial derivative is $[r_0(t)/r_0(y)]^2$. The second checklist condition gives $2/3 \le R \le 4/3$ and $|a - a_0| \le C_F B^2 \beta^2 \omega_T^2/k$. For two trial scalars at norm distance d_η, the corresponding estimate is $|a - \widetilde{a}| \le 4C_F B|\beta| d_\eta \omega_T^2/k$.

Writing the scalar equation as

$$\eta = \beta F_\lambda/F_\lambda(T) + G_T\{U'(\eta) - m^2\eta - 4(a - a_0)\eta'\}$$

gives exactly the source bound $C_N\beta^2$, Lipschitz bound L_N, and norm bounds (20). The first five checklist conditions therefore construct a unique regular scalar in this ball, with positive metric and $a \ge k/2$, uniformly in finite T. The common Bb ball and the invertible contraction linearization give C^1 parameter dependence and $\|\eta_\beta\|_{T,s} \le B$. Differentiating the scalar fixed-point equation gives $|p_\beta - z(T)| \le C_P|\beta|$; its undifferentiated remainder gives $|p - z(T)\beta| \le C_D\beta^2$.

The scalar derivative conditions then give $g_\beta \ge z(T) + w - (C_P + W_3)b - \bar{\delta}F_2/2 \ge w/2$, so the intermediate value theorem constructs its unique root within $|\beta| \le L\delta$. On this root locus the exact constraint gives

$$(a - \sigma/6)(a + \sigma/6) = H^2 - \delta A(\beta) - \delta^2 B_0(\beta).$$

The positive-curvature condition puts the last two terms together in $[a_1\delta/2, 3a_1\delta/2]$. The final two conditions and $2/3 \le R \le 4/3$ give positive residual at $T = 1/k$ and negative residual as $T \to \infty$, for each $\delta > 0$. Sections 3–5 give its strict decrease, hence its unique zero in the tube. Section 7 and Taylor expansion of the exact boundary constraint give Section 8's remainder bounds.

This proof summary uses no shooting data, continuation interpolants, or floating-point residuals. It proves every $0 < \delta \le \bar{\delta}$ satisfying the sufficient inequalities, rather than just selected samples.

10. Registered exact-decimal model: arithmetic corollary

For the specified rational polynomial model, take

$$k = 1/9, \quad W(\eta) = 1/3 + \eta^2 + \eta^3/3, \quad f(\eta) = 1 + c + c\eta, \quad c = \frac{2}{1.0357712571566784} - \frac{4}{3},$$

where the displayed decimal is interpreted exactly as a rational number. Choose

$$r = 1/100, \quad s = 14/9, \quad B = 2, \quad L = c/2 = \frac{48357160712846}{161839258930731}, \quad \bar{\delta} = 1/500, \quad b = L/500.$$

The separate self-contained exact-arithmetic producer `../bounds/evaluate_rescaled_bounds.py` and its output `../bounds/RESCALED_BOUND_CERTIFICATE.json` verify all eleven conditions. They also directly verify polynomial convexity bounds, as redundant checks of the general implication (21). The improved endpoint constant is exactly $C_H^{\mathrm{new}} = 42/5$. The arithmetic gives the rational upper bounds

$$K_\eta^{\mathrm{new}} < 3.196, \qquad K = K_H^{\mathrm{poly}} < 0.254, \qquad \frac{c^2}{384} - K\bar{\delta} > 0.000423. \tag{28}$$

Here the terminating decimals in (28) are also exact rational comparisons, not rounded substitutions into an inequality. Approximate displays are $K_\eta^{\mathrm{new}} \simeq 3.1951835148$, $K \simeq 0.2533589973$; the actual exact fractions are in the certificate.

Consequently, for **every** $0 < \delta \le 0.002$, there is exactly one regular static shell with $T \ge 1/k$ in the stated Dirichlet tube, and

$$\left|\beta + \frac{9c}{64}\delta\right| < 3.196\delta^2,$$

$$\left|H^2 - \frac{1+c}{27}\delta - \left[\frac{(1+c)^2}{36} - \frac{c^2}{384}\right]\delta^2\right| < 0.254\delta^3. \tag{29}$$

At the largest detuning these bounds are respectively 1.2784×10^{-5} and 2.032×10^{-9}. Relative to the distinct model with scalar-independent detuning $f_{\mathrm{metric}}(\eta) = 1 + c$, its exactly known curvature is

$$H^2_{\mathrm{metric}} = \frac{1+c}{27}\delta + \frac{(1+c)^2}{36}\delta^2,$$

and (28) proves the strict statement

$$H^2 - H^2_{\mathrm{metric}} < -0.000423\delta^2 < 0 \qquad (0 < \delta \le 0.002). \tag{30}$$

This corollary covers the three previously investigated detunings 0.0005, 0.001, and 0.002, and all positive detunings below them in the displayed interval. It certifies the mathematical branch and its analytic error bound; it does not certify the floating-point shooting values individually or identify their parameter-rounding conventions with the exact-decimal model.

All uniqueness conclusions remain restricted to the constructed tube. The proof neither rules out other large-field interiors nor analyzes perturbative stability. Exact arithmetic evaluations and independent internal review are separate artifacts; they do not establish external peer review or novelty.